

Final Mock Exam — Set C — Solutions

Lectures 1–12 (comprehensive)

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Overview of the final mock exam — Set C

Problem	Topic	Lecture slides	Points
P1	Moving beyond linearity — splines	Ch. 7 — Moving Beyond Linearity (Lecture 9)	16
P2	Generalised additive models	Ch. 7 — Moving Beyond Linearity (Lecture 9)	8
P3	Trees by hand (regression & classification)	Ch. 8 — Tree-Based Methods (Lecture 10)	20
P4	Ensembles — bagging, RF, boosting	Ch. 8 — Tree-Based Methods (Lecture 10)	16
P5	Neural networks by hand	Ch. 10 — Deep Learning (Lecture 11)	20
P6	Multiple testing	Ch. 13 — Multiple Testing (Lecture 12)	20
P7	Cumulative essentials	Ch. 2–6 (Lectures 1–8)	20
Σ			120

Exam conditions

120 minutes for 120 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 — task

Problem 1 (16 P) — Moving beyond linearity

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) [4 P] Degrees of freedom of a cubic spline with K interior knots — derive by counting parameters and constraints; evaluate for $K = 3$.
- (b) [4 P] Degrees of freedom of a *natural* cubic spline with the same $K = 3$ knots; which extra constraint acts where, and why is it useful?
- (c) [4 P] Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.
- (d) [4 P] Truncated-power basis of a cubic spline with one knot ξ ; define $(x - \xi)_+^3$; why does adding it keep the fit smooth at ξ ?

Problem 1 — solution (a) and (b)

Solution P1 — (a) and (b)

(a) K knots $\Rightarrow K + 1$ cubic pieces $\times 4$ coefficients $= 4K + 4$ parameters; continuity of g and g' and g'' at each knot $\Rightarrow 3K$ constraints:

$$\text{df} = 4(K + 1) - 3K = K + 4 \quad \Rightarrow \quad K = 3 : \text{df} = 7.$$

(Equivalently: basis $1, x, x^2, x^3$ plus one truncated power per knot.)

(b) Natural spline: *linear beyond the boundary knots* (regions $X < \xi_1$ and $X > \xi_K$); killing the quadratic and cubic terms there costs 2 parameters per boundary region:

$$\text{df} = (K + 4) - 4 = K \quad \Rightarrow \quad K = 3 : \text{df} = 3.$$

Why useful: splines are most erratic near the boundaries where data are sparse; the linearity constraint stabilises the fit there (lower variance for a little extra bias).

Problem 1 — solution (c) and (d)

Solution P1 — (c) and (d)

(c) RSS term = fidelity to the data; $\lambda \int g''(t)^2 dt$ penalises curvature (wiggleness); λ trades the two off.

- $\lambda \rightarrow 0$: no penalty $\Rightarrow \hat{g}$ *interpolates* the training points exactly — extremely wiggly, zero training RSS, severe overfitting (effective df $\rightarrow n$).
- $\lambda \rightarrow \infty$: curvature infinitely expensive $\Rightarrow g'' \equiv 0 \Rightarrow \hat{g}$ is the *least-squares straight line* (effective df $\rightarrow 2$).

λ (chosen by LOOCV) steers the effective df smoothly between n and 2.

(d) $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$ with

$$(x - \xi)_+^3 = \begin{cases} (x - \xi)^3 & x > \xi \\ 0 & x \leq \xi \end{cases}$$

$(x - \xi)_+^3$ has value and first and second derivative 0 at $\xi \Rightarrow$ only the third derivative jumps: the fit stays continuous with continuous g' and g'' — a cubic spline.

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) [3 P] General form of a GAM for quantitative Y ; how does it generalise multiple linear regression?
- (b) [3 P] One advantage concerning the interpretation of the f_j ; the main structural limitation — and its repair.
- (c) [2 P] Backfitting in one or two sentences.

Problem 2 — solution

Solution P2 — GAMs

(a) $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \varepsilon$.

Linear regression is the special case $f_j(X_j) = \beta_j X_j$; a GAM replaces each linear term by a smooth non-linear f_j (spline, local regression) but keeps the *additive* structure.

(b) *Advantage*: additivity $\Rightarrow$ the effect of each X_j is its own function f_j , which can be plotted and interpreted individually while the other predictors are held fixed — one panel per predictor, as readable as a coefficient table.

Limitation: no *interactions* — the effect of X_j cannot depend on X_k .

Repair: add interaction components manually, e.g. $X_j \times X_k$ or a two-dimensional smooth $f_{jk}(X_j, X_k)$.

(c) *Backfitting*: cycle through $j = 1, \dots, p$ and re-fit each f_j to the partial residuals $r_i = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$ with all other functions held fixed; repeat until convergence.

Problem 3 — task

Problem 3 (20 P) — Trees by hand

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Delivery vans x and parcels delivered y (1,000) for $n = 6$ depots; one split at s : $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i (vans)	1	2	3	4	5	6
y_i (parcels)	3	5	4	9	11	10

(a) **[8 P]** Evaluate the cutpoints $s = 2.5$ and $s = 3.5$ by total RSS; which is chosen? (b) **[2 P]** Prediction for a new depot with $x = 5$ vans. (c) **[6 P]** Node with 5 “Electronics” / 3 “Clothing” / 1 “Books”: proportions; Gini; entropy ($\ln \frac{5}{9} = -0.5878$; $\ln \frac{1}{3} = -1.0986$; $\ln \frac{1}{9} = -2.1972$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Problem 3 — solution (a) and (b)

Solution P3 — (a) and (b)

(a) Region means and RSS per cutpoint ($\bar{y} = 42/6 = 7$; root RSS = 58):

s	R_1 (y-values)	$\hat{y}_{R_1}$	R_2 (y-values)	$\hat{y}_{R_2}$	total RSS
2.5	{3,5}	4	{4,9,11,10}	8.5	$2 + 29 = 31$
3.5	{3,5,4}	4	{9,11,10}	10	$2 + 2 = 4$

Details: $s = 2.5$: $(3 - 4)^2 + (5 - 4)^2 = 2$; $(4 - 8.5)^2 + (9 - 8.5)^2 + (11 - 8.5)^2 + (10 - 8.5)^2 = 29$.

$s = 3.5$: $(3 - 4)^2 + (5 - 4)^2 + (4 - 4)^2 = 2$; $(9 - 10)^2 + (11 - 10)^2 + (10 - 10)^2 = 2$.

Greedy recursive binary splitting picks the smaller RSS: $s = 3.5$ ($4 \ll 31$; in fact the best of all five cutpoints) — it isolates the three high-volume depots.

(b) $x = 5 \geq 3.5 \Rightarrow$ region $R_2 \Rightarrow \hat{y} = \hat{y}_{R_2} = 10$ (i.e. 10,000 parcels).

Problem 3 — solution (c) and (d)

Solution P3 — (c) and (d)

(c) $\hat{p}_E = 5/9$; $\hat{p}_C = 3/9 = 1/3$; $\hat{p}_B = 1/9$.

$$G = \frac{5}{9} \cdot \frac{4}{9} + \frac{3}{9} \cdot \frac{6}{9} + \frac{1}{9} \cdot \frac{8}{9} = \frac{20+18+8}{81} = \frac{46}{81} = 0.568$$

$$D = -\left(\frac{5}{9} \ln \frac{5}{9} + \frac{1}{3} \ln \frac{1}{3} + \frac{1}{9} \ln \frac{1}{9}\right) = 0.3266 + 0.3662 + 0.2441 = 0.937$$

Predicted class: majority = **Electronics** (estimated probability $5/9 \approx 0.556$). Both measures are 0 for a pure node — this node is quite impure (no class dominates).

(d) α = price per terminal node $|T|$:

- $\alpha = 0$: criterion = training RSS $\Rightarrow$ full tree T_0 is optimal.
- $\alpha \uparrow$: leaves must pay for themselves; branches with small RSS gain are pruned $\Rightarrow$ nested sequence of ever smaller subtrees (weakest-link pruning).
- Choice: *cross-validation* over α ; refit the chosen subtree on all data — balances bias against variance.

Problem 4 — task

Problem 4 (16 P) — Ensembles

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

- (a) [6 P] B identically distributed trees with variance σ^2 and pairwise correlation ρ :
 $\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 50$ and $\rho = 0.4$ and $\sigma^2 = 16$; compare with one tree. (iii) Limit as $B \rightarrow \infty$?
- (b) [4 P] Random forests: modification at each split; recommended m for *regression* with $p = 30$; why does it improve on bagging (use the formula)?
- (c) [4 P] Boosting: the three tuning parameters; the slow-learning idea.
- (d) [2 P] What does the OOB error estimate and why is it free?

Problem 4 — solution (a) and (b)

Solution P4 — (a) and (b)

(a) (i) Averaging drives the second term $\frac{1-\rho}{B}\sigma^2 \rightarrow 0$ (the independent part of the trees' variability); the first term $\rho\sigma^2$ is independent of B — correlation between bootstrap trees limits the reduction.

(ii) $\text{Var} = 0.4 \cdot 16 + \frac{0.6}{50} \cdot 16 = 6.4 + 0.192 = 6.592$ vs. $\sigma^2 = 16$ for a single tree ($\approx 58.8\%$ less).

(iii) $B \rightarrow \infty$: variance $\rightarrow \rho\sigma^2 = 6.4$ (floor; more trees never overfit, they only stabilise).

(b) At *each split* only a random subset of m of the p predictors may be used; regression default $m \approx p/3$
 $\Rightarrow p = 30$: $m = 30/3 = 10$.

With bagging a dominant predictor tops nearly every tree $\Rightarrow$ trees alike $\Rightarrow \rho$ high. Restricting candidates *decorrelates* the trees ($\rho \downarrow$), which lowers the floor $\rho\sigma^2$ — the part of the variance that averaging alone cannot remove.

Problem 4 — solution (c) and (d)

Solution P4 — (c) and (d)

(c) Boosting for regression trees:

- B — **number of trees**: added sequentially; too many $\Rightarrow$ slow overfitting; choose by CV.
- λ — **shrinkage** (e.g. 0.01 or 0.001): scales each new tree's contribution; smaller $\lambda \Rightarrow$ slower learning $\Rightarrow$ larger B needed.
- d — **splits per tree** (interaction depth): complexity and interaction order of each tree; often stumps ($d = 1$) suffice (additive model).

Slow learning: each tree is fitted to the current *residuals* and only the fraction λ is added — the model improves gradually exactly where it still errs, instead of fitting the data hard at once; this tends to generalise better.

(d) Each bootstrap sample misses $\approx 1/3$ of the observations (out-of-bag); predicting every observation only with trees that never saw it yields the **OOB error** — a valid *test-error* estimate, free as a by-product of the bootstrap (no validation set or CV loop needed).

Problem 5 — task

Problem 5 (20 P) — Neural networks by hand

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Input $x = (2,3)$; hidden ReLU units $g(z) = \max\{0,z\}$; linear output:

$$A_1 = g(-3 + 2x_1 + 1x_2) \quad A_2 = g(2 - 2x_1 - 1x_2) \quad f(x) = -2 + 3A_1 + 2A_2$$

- (a) **[8 P]** Forward pass step by step: pre-activations; A_1 and A_2 ; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?
- (b) **[6 P]** Logits $z = (2.0, 0.0, -1.0)$; true class 2: all three softmax probabilities and the cross-entropy loss ($e^2 = 7.389$; $e^0 = 1.000$; $e^{-1} = 0.368$; $\ln 8.757 = 2.170$).
- (c) **[4 P]** Parameter count of a $p = 6 \rightarrow k = 5 \rightarrow K = 3$ network with biases (show both layers).
- (d) **[2 P]** Conv layer: 64×64 input; $F = 3$; $P = 1$; $S = 2$: output size via $\lfloor (W - F + 2P)/S \rfloor + 1$.

Problem 5 — solution (a) forward pass

Solution P5 — (a) forward pass

Pre-activations:

$$z_1 = -3 + 2 \cdot 2 + 1 \cdot 3 = 4$$

$$z_2 = 2 - 2 \cdot 2 - 1 \cdot 3 = -5$$

ReLU activations:

$$A_1 = \max\{0, 4\} = 4$$

$$A_2 = \max\{0, -5\} = 0$$

Output:

$$f(x) = -2 + 3 \cdot 4 + 2 \cdot 0 = 10$$

The ReLU *switched unit 2 off*: its negative pre-activation is clipped to 0, so it contributes nothing for this input (though it can for others).

Why essential: with a linear g the whole network collapses into one linear function of x regardless of depth — the non-linearity is what lets hidden layers build flexible functions (piecewise-linear surfaces with ReLU).

Problem 5 — solution (b) softmax and cross-entropy

Solution P5 — (b) softmax and cross-entropy

Exponentials and their sum:

$$e^{2.0} = 7.389 \quad e^{0.0} = 1.000 \quad e^{-1.0} = 0.368 \quad \Rightarrow \quad 7.389 + 1.000 + 0.368 = 8.757$$

Softmax probabilities:

$$\hat{p}_1 = \frac{7.389}{8.757} = 0.844 \quad \hat{p}_2 = \frac{1.000}{8.757} = 0.114 \quad \hat{p}_3 = \frac{0.368}{8.757} = 0.042$$

(sum = 1.000 ✓). Cross-entropy for the *true* class 2:

$$-\ln \hat{p}_2 = -\ln \frac{1.000}{8.757} = -(0 - \ln 8.757) = \ln 8.757 = 2.170$$

The model's most probable class is class 1 ($\hat{p}_1 = 0.844$), but the loss is evaluated at the true class 2 (only 0.114) — hence the sizeable loss; a perfect $\hat{p}_2 = 1$ gives 0.

Problem 5 — solution (c) and (d)

Solution P5 — (c) parameters and (d) convolution

(c) Fully connected $6 \rightarrow 5 \rightarrow 3$ with biases:

Layer	Weights	Biases	Parameters
input $\rightarrow$ hidden	$6 \cdot 5 = 30$	5	35
hidden $\rightarrow$ output	$5 \cdot 3 = 15$	3	18
total			53

(d) Output width = $\left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{64 - 3 + 2 \cdot 1}{2} \right\rfloor + 1 = \lfloor 31.5 \rfloor + 1 = 32 \Rightarrow$ output feature map 32×32 : stride $S = 2$ roughly halves the 64×64 input (the floor discards the fractional part before adding 1).

Problem 6 — task

Problem 6 (20 P) — Multiple testing

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

- (a) **[4 P]** $m = 40$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER $((0.95)^{40} = 0.1285)$; interpret.
- (b) **[7 P]** Six ordered p -values 0.001; 0.008; 0.011; 0.030; 0.041; 0.260 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?
- (c) **[6 P]** Benjamini–Hochberg at $q = 0.05$ on the same p -values: thresholds jq/m ; rejections; what exactly does FDR control guarantee?
- (d) **[3 P]** Confirmatory phase-III trial with $m = 5$ pre-specified endpoints (one false positive $\Rightarrow$ an ineffective drug approved): FWER or FDR? Justify.

Problem 6 — solution (a) FWER

Solution P6 — (a) FWER for 40 independent tests

$\text{FWER} = \Pr(V \geq 1)$ — the probability of *at least one* false rejection among the m tests (V = number of falsely rejected true nulls).

With all 40 nulls true and independent tests at level 0.05:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{40} = 1 - (0.95)^{40} = 1 - 0.1285 = 0.8715$$

Interpretation: although each test individually errs with probability 5% only, the chance of at least one spurious “discovery” among 40 unadjusted tests is about 87% — multiple testing without adjustment almost guarantees false findings.

Problem 6 — solution (b) Bonferroni and Holm

Solution P6 — (b) Bonferroni and Holm at level 0.05

Bonferroni: reject iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$: only 0.001 and 0.008 pass ($0.011 > 0.00833$) $\Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$: **2 rejections**.

Holm (step-down): compare $p_{(j)}$ with $\alpha/(m - j + 1)$; stop at first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
$\alpha/(m - j + 1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	stop	—	—

$0.011 \leq 0.0125$ but $0.030 > 0.0167 \Rightarrow$ reject $H_{(1)}$ and $H_{(2)}$ and $H_{(3)}$: **3 rejections**.

Holm $\supseteq$ Bonferroni: Holm's hurdles $\alpha/(m - j + 1) \geq \alpha/m$ for all j (equality only at $j = 1$) $\Rightarrow$ uniformly more powerful — and it still controls the FWER without independence assumptions. Here Holm rejects one more ($H_{(3)}$).

Problem 6 — solution (c) and (d)

Solution P6 — (c) Benjamini–Hochberg and (d) FDR vs FWER

(c) Thresholds $jq/m = j \cdot 0.05/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \leq jq/m?$	yes	yes	yes	yes	yes	no

Largest such j : $L = 5$ ($0.041 \leq 0.0417$; note $0.260 > 0.0500$) $\Rightarrow$ reject *all* of $H_{(1)}$ to $H_{(5)}$: **5 rejections** (most liberal: 5 vs. Holm 3, Bonferroni 2). *Guarantee*: $\text{FDR} = E[V / \max(R, 1)] \leq q$ — *on average* at most 5% of rejected hypotheses are false discoveries (no bound on $\Pr(V \geq 1)$; that is FWER).

(d) Control the **FWER**. A confirmatory trial with only $m = 5$ pre-specified endpoints, where a single false positive is very costly (an ineffective drug approved), demands a bound on the probability of *any* false rejection — exactly FWER control (e.g. Holm at 0.05). FDR only limits the *expected proportion* of false endpoints and would tolerate an occasional false positive: fine for exploratory screens, not here.

Problem 7 — task

Problem 7 (20 P) — Cumulative essentials

Lecture slides: Chapters 2–6 (Lectures 1–8).

- (a) **[4 P]** Bootstrap: (i) show $\Pr(\text{obs. not in sample}) = (1 - 1/n)^n$ and its limit ($e^{-1} = 0.368$); (ii) fraction included as $n \rightarrow \infty$; (iii) exact inclusion probability for $n = 5$ ($0.8^5 = 0.328$).
- (b) **[4 P]** Bias–variance at x_0 : method (1) $\text{Var} = 3.5$, $\text{Bias} = -1.0$, $\text{Var}(\varepsilon) = 2.0$; method (2) $\text{Var} = 0.5$, $\text{Bias} = -2.5$, $\text{Var}(\varepsilon) = 2.0$. Expected test MSE of each; which to deploy; lowest achievable MSE?
- (c) **[3 P]** Elastic-net penalty: what does it inherit from lasso and from ridge? When does it clearly beat the lasso?
- (d) **[3 P]** Collinearity (corr. 0.95): effect on OLS estimates; $\text{VIF}_j = 1/(1 - R_j^2)$ for $R_j^2 = 0.9$ ($\sqrt{10} = 3.162$).
- (e) **[6 P]** Match to {linear regression; lasso; logistic regression; random forest}, each once: (i) Δwage per school-year with CI ($n = 3,000$); (ii) biomarker from 8,000 metabolites ($n = 80$) with a short list; (iii) churn yes/no with odds for management; (iv) electricity demand from hundreds of interacting variables, accuracy only.

Problem 7 — solution (a) to (c)

Solution P7 — (a) to (c)

(a) Each draw misses a fixed observation with prob. $1 - \frac{1}{n}$; all n draws (independent) miss it with prob. $(1 - \frac{1}{n})^n \rightarrow e^{-1} = 0.368$. So it is *included* with prob. $1 - 0.368 = 0.632$ ($\approx 63.2\%$ of observations appear; the rest are OOB). For $n = 5$: $1 - 0.8^5 = 1 - 0.328 = 0.672$.

(b) $\text{MSE} = \text{Var} + \text{Bias}^2 + \text{Var}(\varepsilon)$:

$$\text{MSE}_1 = 3.5 + (-1.0)^2 + 2.0 = 6.5 \quad \text{MSE}_2 = 0.5 + (-2.5)^2 + 2.0 = 8.75$$

Deploy the **flexible spline** (method 1, $6.5 < 8.75$): its higher variance is more than offset by its much smaller squared bias. Lowest achievable $\text{MSE} = \text{irreducible error } \text{Var}(\varepsilon) = 2.0$.

(c) Elastic net: $\lambda \left[\frac{1-\alpha}{2} \sum_j \beta_j^2 + \alpha \sum_j |\beta_j| \right]$. From *lasso* (ℓ_1): exact zeros / variable selection; from *ridge* (ℓ_2): graceful handling of correlated predictors and selecting $> n$ variables when $p > n$. It beats the lasso with *groups of correlated predictors* (lasso picks one and drops the rest; elastic net keeps the group — “grouping effect”).

Problem 7 — solution (d) and (e)

Solution P7 — (d) and (e)

(d) Collinearity does not bias OLS but makes it *imprecise/unstable*: inflated variances and standard errors, wide CIs, coefficients that can flip sign. With $R_j^2 = 0.9$, $VIF_j = 1/(1 - 0.9) = 10$, so $\text{Var}(\hat{\beta}_j)$ is $10\times$ larger and its SE is inflated by $\sqrt{10} = 3.162$ (rule of thumb: $VIF > 5-10$ is problematic).

(e) Matching (each method once):

- (i) **Linear regression** — quantitative response; *inference* (effect size with CI) from OLS standard errors.
- (ii) **Lasso** — $p = 8,000 \gg n = 80$: OLS infeasible; ℓ_1 selects a short list of metabolites.
- (iii) **Logistic regression** — binary response; coefficients give odds ratios $e^{\hat{\beta}_j}$: interpretable for management.
- (iv) **Random forest** — automatic non-linearities and interactions; strong accuracy with many predictors; interpretability not needed.